

# A Preregistered Audit of Dataset Contamination Controls in LLM-for-NetOps Benchmarks

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**Abstract**—We preregistered and executed a paired, confirmatory audit to test whether conservative deterministic contamination detectors (exact-match + fuzzy 5-gram + provenance heuristics) flag items in a reproducible 150-item NetOps sample and whether applying preregistered contamination controls changes validator-based success rates. Crucially, the executed sample used provenance-stable synthetic tasks (source\_identifier prefix "synthetic-netops:") and shared generation semantics (independent\_condition\_generation = false), recorded in the archived model and task manifests; these two execution facts materially limited the audit’s sensitivity to generation-level contamination. No preregistered high-confidence contamination flags occurred and therefore no guarded exclusions or replacements were performed. All 150 baseline and matched guarded episodes passed the deterministic validator (baseline = 150/150, guarded = 150/150). The paired success-rate difference = 0.0 percentage points (95% CI [0.0, 0.0]); exact McNemar two-sided  $p = 1.0$ . We reconcile the preregistration→execution gap with artifact-grounding diagnosis, explain why the run is degenerate for detecting public-benchmark leakage, and provide concrete, artifact-anchored recommendations for audit designs that would be sensitive to contamination in web-sourced benchmarks.

## I. INTRODUCTION

**Motivation.** Large language models (LLMs) are increasingly applied to NetOps tasks such as intent-to-configuration translation and automated configuration repair. Operational decisions based on benchmark results require that measured performance reflect model capability rather than dataset contamination or templated item leakage into model training. Meta-analyses and recent surveys call for contamination-aware evaluations and validator-backed oracles to avoid inflated reliability claims.

**Research question and contribution.** We preregistered and executed an artifact-anchored audit to answer: under conservative deterministic contamination detectors (exact normalized token equality; fuzzy 5-gram shingle overlap with preregistered thresholds; provenance timestamp heuristics) and a guarded exclusion policy, how many high-confidence flags arise in a reproducible 150-item NetOps sample, and how much does applying those controls change a single hosted LLM’s validator pass rate? Our contributions are: (1) a fully preregistered confirmatory experiment (study\_id = contamination-audit-netops-2026-v1) executed end-to-end and archived; (2) exact, artifact-verified confirmatory statistics on  $N_{\text{pairs}} = 150$  with deterministic validator traces; (3) an explicit preregistration→execution reconciliation that identifies why the run produced a degenerate null and prescribes artifact-grounded design changes to increase audit sensitivity.

**Scope and bounded claims.** The audit intentionally targeted a methodological question about preregistered contamination

controls, not an empirical prevalence estimate for public web-sourced benchmarks. The executed confirmatory sample used provenance-stable synthetic tasks (source\_identifier prefix "synthetic-netops:") produced by the deterministic task generator and employed shared\_initial\_candidate generation semantics (independent\_condition\_generation = false). Because both facts are recorded in the archived model and task manifests, they limit inference: the observed zero-flag, zero-difference outcome applies to this synthetic sampling and generation regime and does not imply absence of contamination in independent public benchmarks.

## II. RELATED WORK

**LLM-driven NetOps systems and validators.** Prior system and benchmark papers demonstrate that LLMs can translate intent into configuration artifacts and can propose repairs; integrating verifiers or iterative validate–repair loops improves average repair quality compared to naive generation. Formal network verification tools (e.g., Header Space Analysis and localized subspecification) are practical oracles for reachability and policy correctness and are commonly used in generate–validate pipelines.

**Evaluation-validity critiques and contamination audits.** Surveys and meta-analyses highlight two recurring evaluation risks: (a) contamination—benchmark items or templated prompts appearing in model training data can inflate measured accuracy; and (b) construct invalidity—benchmarks dominated by low-complexity or templated tasks may not represent operational difficulty. Recommendations in the literature include deterministic fingerprinting, fuzzy n-gram overlap, provenance tracing, and reporting of detector score distributions. Our audit implements a conservative subset of these defenses and emphasizes an artifact-anchored preregistration→execution reconciliation.

**Positioning.** Unlike broad capability benchmarks that measure absolute performance across model families, this study evaluates whether a specific preregistered set of contamination controls would alter validator-measured success on a reproducible sample under the archived execution semantics. Where prior works focus on detector design or end-to-end agentic repair effectiveness, our contribution is a small, fully preregistered audit and a tight artifact-grounded diagnosis of why the experimental choices produced a degenerate result.

### III. METHODOLOGY

Preregistration and primary estimand (design freeze). The study was preregistered under study\_id = contamination-audit-netops-2026-v1 and froze the confirmatory design (estimands, sampling, detectors, exclusion/replacement rules, and analysis procedures) prior to observing model outputs. The primary estimand is the paired success-rate difference (guarded - baseline) across item $\times$ model pairs; the preregistered confirmatory test is an exact McNemar two-sided test and paired bootstrap percentile 95% CI computed with 10,000 resamples and seed = 1729. The registered analysis executor is paired\_binary\_analysis\_v1 and the preregistration explicitly specified sensitivity runs (alternate fuzzy thresholds and failure-as-zero handling) and a multiplicity plan that treats the primary test as the single confirmatory inference.

Contamination detectors and guarded policy (operationalized). The audit implemented three deterministic detectors as preregistered defenses: (1) exact normalized token equality after canonicalization (whitespace and token normalization); (2) fuzzy 5-gram shingle overlap (Jaccard/overlap) computed on 5-gram token shingles with preregistered high-confidence threshold  $\geq 0.80$  and medium band  $[0.50, 0.80)$ ; and (3) provenance timestamp heuristics that compare item source timestamps/DOIs against the model public-cutoff metadata. The detection pipeline was implemented to produce high-confidence flags when exact equality OR 5-gram overlap  $\geq 0.80$  hold; medium flags were marked for overlap in  $[0.50, 0.80)$  or provenance risk. The preregistered guarded policy deterministically excludes only high-confidence flagged items and replaces them with deterministic retemplated + topology-augmented items sampled from the same generator using fixed replacement seeds; replacements were required to satisfy an embedding-similarity constraint (preregistered cosine threshold  $\geq 0.85$ ) and to pass functional sanity checks before being accepted. The detector implementation persisted flagged events and replacement traces by design; the code path for detailed per-item overlap table persistence was activated only when threshold crossings or replacements occurred (see Results for the artifact consequence).

Sampling, task generation, and generation semantics (preregistered options). The preregistration allowed sampling from a specified benchmark scope; for reproducibility, the design permitted using the deterministic task generator (deterministic\_netops\_task\_generator\_v5) with fixed generation seeds and specified source identifiers. The preregistered execution\_contract allowed independent\_condition\_generation (two independent model calls per item, producing two independent outputs) or shared\_initial\_candidate semantics (single shared generation reused across conditions) depending on experimental constraints. The confirmatory sample target was task\_count = 150 items (planned\_episode\_count = 300 episodes if independent generations were used). The sampling plan included deterministic replacement rules for excluded items and explicit seeds for replacement sampling so that paired alignment across baseline and guarded conditions is preserved in the event of

exclusions.

Model and provider configuration (declared). The execution contract specified a single hosted model family for the confirmatory run (declared\_model\_version = "gpt-5-mini"). The model configuration used in the pipeline (documented in execution/model\_configuration.json in the archive) set provider = openai\_responses, max\_output\_tokens = 2000, maximum\_attempts\_per\_call = 1, reasoning\_effort = "minimal", and store = false. The preregistration capped maximum\_model\_calls = 300 (one per condition per item under independent generation), but allowed shared-generation runs to consume fewer calls when allowed by the design.

Deterministic NetOps verifier and scoring rule (oracle). The verification oracle is deterministic\_netops\_validator\_v5 (version 5.0). The validator follows a parse  $\rightarrow$  state-update  $\rightarrow$  property-check pipeline: it parses candidate configuration commands into normalized command sequences, simulates the command sequence against the canonical initial state to produce an expected final state, and evaluates a set of preregistered workflow and invariant constraints (e.g., required command counts, ordered shutdown/restore patterns, protected interfaces, group availability constraints). The preregistered scoring rule is full\_intent\_constraint\_validation: a binary pass requires reaching the intended final state while respecting all workflow constraints; the validator also records diagnostic fields per episode (normalized\_configuration, parsed\_command\_count, required\_command\_count, observed\_touched\_field\_count, violation\_codes) and archives detailed traces for reproducibility. These trace fields form the deterministic basis for the paired binary outcomes used by the paired analysis executor.

Missingness, retries, and analysis handling (preregistered). The preregistered missingness plan mandated a single automated retry for model/API call failures within a five-minute window; if a retry failed, the analysis default was 'complete\_pair\_primary' (the pair treated per the analysis executor rules) with a sensitivity run treating failed calls as failures (zero). Exclusion rules were deterministic and applied only to high-confidence flagged items; replacements were deterministic and seeded. The preregistration included sensitivity analyses for fuzzy thresholds at 0.50 and 0.65 and subgroup analyses by templating detector output.

Planned power and interpretation guidance (design constraints). The preregistration documented power guidance acknowledging that a two-generation per item design (independent\_condition\_generation = true) or a multi-model plan materially increases the number of independent trials and therefore the power to detect modest paired effects (target detect  $\approx 5$  percentage points with 80% power under conservative assumptions). In contrast, shared\_generation semantics (single shared model call per item) reduce the number of independent model calls and therefore lower sensitivity to generation-level contamination unless deterministic high-confidence flags occur. These planning notes were preregistered to guide interpretation of null or degenerate paired outcomes.

Reproducibility, archival, and artifact paths. All sampling

seeds, generator identifiers, validator implementations, detector code, and analysis executors were recorded and archived as part of the execution bundle. Key artifact paths produced by the pipeline include `execution/task_manifest.jsonl`, `execution/model_configuration.json`, `execution/raw_results.jsonl`, `analysis/paired_contingency_table.csv`, and `analysis/contamination_summary.csv`; the full execution manifest contains hashes and a complete artifact index. The preregistration SHA and the manifest entries are archived and available for verification in the execution bundle. The analysis executor parameters (bootstrap resamples = 10000, bootstrap\_seed = 1729) and the exact paired\_test method (exact\_mcnemar\_two\_sided) were recorded in the analysis log and are reproduced in the archived analysis artifacts.

#### IV. RESULTS

Execution accounting and reconciliation. The archived execution manifest records `planned_episode_count` = 300 (150 items  $\times$  2 conditions) and `completed_episode_count` = 300 with `failed_episode_count` = 0. Because the run used `shared_initial_candidate` semantics and no high-confidence contamination exclusions occurred, the pipeline consumed `model_calls_used` = 150 (one shared model generation per item) rather than two independent generations per condition; provider call auditing corroborates `hosted_model_call_event_count` = 150, `cache_miss_count` = 150 and `cross_condition_cache_key_reuse_observed` = false. The deterministic\_reconciliation artifact confirms `complete_pair_count` = 150 and internal accounting consistency (`n_11` = 150, `n_10` = `n_01` = `n_00` = 0).

Contamination detectors and archived evidence of absence. Under the preregistered high-confidence rule (exact match OR fuzzy 5-gram overlap  $\geq$  0.80) the detector workflow produced zero high-confidence flags for the executed sample. The analysis artifact `contamination_summary.csv` is empty (no flagged rows), and the run therefore performed no deterministic exclusions or replacements. Per the preregistered detector implementation, detailed per-item fuzzy overlap scores and replacement traces are persisted only when threshold crossings or replacements occur; because no items met the high-confidence threshold, the archive contains the detector’s flag events (none) but not a full per-item overlap table. We therefore report the artifact-grounded null finding (no high-confidence flags) and emphasize that the absence of persisted overlap summaries is itself an execution artifact with diagnostic meaning (see below).

Validator outcomes: concordant passes and formal statistics. The deterministic\_netops\_validator\_v5 returned binary pass for every baseline episode (`baseline_success_count` = 150/150, `baseline_success_rate` = 1.0) and for every guarded episode (`guarded_success_count` = 150/150, `guarded_success_rate` = 1.0). The paired contingency table is `n11` = 150, `n10` = 0, `n01` = 0, `n00` = 0. The primary estimand (`paired_success_rate_difference_guarded_minus_baseline`) = 0.0 percentage points. The exact McNemar two-sided test on these counts yields  $p$  = 1.0. The preregistered paired

bootstrap percentile CI (10,000 resamples, seed = 1729) is [0.0, 0.0]. The zero-width CI and  $p$  = 1.0 are a direct consequence of the absence of any discordant pairs; these are artifact-verified outcomes derived from the archived `paired_contingency_table.csv` and `condition_summary.csv`.

Representative diagnostic episodes. Representative archived episode traces illustrate validator behavior and the nature of shared generations. For example, pair-000001 (task-000001) has `normalized_response` identical to the reference, `parsed_command_count` = 4 and `required_command_count` = 4, `observed_touched_field_count` = 3, and validator verdict = `VALID_CONFIGURATION`. Pair-000002 (task-000002) shows `parsed_command_count` = 2 (`required_command_count` = 2, `level` = hard) and validator verdict = `VALID_CONFIGURATION`. Pair-000003 (task-000003) shows `parsed_command_count` = 6 (`required_command_count` = 6, `level` = hard) and validator verdict = `VALID_CONFIGURATION`. These exemplar traces (archived per-episode validator\_trace records) demonstrate that when the model output aligns with the preregistered intent and workflow constraints the validator produces a detailed, machine-readable confirmation; in this execution these per-episode confirmations are uniformly positive.

Why the paired test was uninformative for generation-level contamination. Two artifact-recorded execution facts explain the degenerate concordance. First, the confirmatory sample comprised provenance-stable, deterministically generated synthetic items (source\_identifier prefix "synthetic-netops:"), so exact canonicalized matches and very high fuzzy 5-gram overlaps to public corpus fingerprints were unlikely a priori. Second, `independent_condition_generation` = false (`shared_initial_candidate`) was in effect: when no high-confidence flags triggered exclusions the same single model output was reused for both baseline and guarded conditions. Together these facts guarantee that, absent deterministic exclusions, baseline and guarded episodes for each item are identical in content and validator outcome—producing `n11` = 150 and zero discordance by construction. The execution manifest and model\_configuration artifact explicitly encode these settings and thus ground this diagnosis in archived evidence rather than post hoc speculation.

Detector persistence policy and the missing overlap distribution. The preregistered detector persisted detailed fuzzy overlap scores only when an item crossed a threshold or when replacements were enacted; the archived absence of persisted per-item overlap scores therefore reflects the detector’s control flow under the executed conditions (no threshold crossings). This design choice reduced the post-hoc ability to report summary overlap statistics (min/median/percentiles) for the executed sample from the archive. The missing overlap distribution is an execution artifact: it indicates that no items came close enough to the high-confidence threshold to invoke persistence logic, but it does not reveal whether many items would have had moderate overlaps below the threshold (the archive contains no full table of low-confidence overlaps). This factual gap motivates our concrete recommendation to persist

per-item overlap scores regardless of threshold crossings in future audits to enable transparent reporting of the overlap distribution and sensitivity analyses.

Sensitivity to preregistered design choices. The preregistration explicitly documented alternative, higher-sensitivity designs: (a) `independent_condition_generation = true` so baseline and guarded generations are independent (doubling model calls to 300) and allowing per-item discordance to appear even without deterministic flagging; (b) multi-model replication to reveal model-specific memorization signals; and (c) lower fuzzy thresholds or paraphrase-robust detectors plus human adjudication. The archived power guidance in the preregistration explains that a 300 independent-call design or a multi-model plan would materially lower the detectable paired difference (planned target  $\approx 5$  percentage points at 80% power under conservative assumptions). Because the executed run used the lower-sensitivity shared-generation regime and synthetic sampling, the artifact-grounded conclusion is limited to this regime: deterministic detectors did not trigger exclusions and validators returned universal passes for the shared outputs.

Summary of archival provenance supporting the numerical results. All numeric summaries above—`completed_episode_count = 300`, `model_calls_used = 150`, `baseline_success_count = 150`, `guarded_success_count = 150`, `n11 = 150`, paired bootstrap CI [0.0,0.0] (10,000 resamples, seed = 1729), and McNemar  $p = 1.0$ —are computed by the preregistered analysis executor and are recorded in `analysis/results.json` and in the small CSV artifacts `condition_summary.csv` and `paired_contingency_table.csv`. The `contamination_summary.csv` artifact contains no flagged rows for this execution. Per-episode raw responses and `validator_trace` objects are archived for independent verification of every binary outcome used in the paired analysis.

## V. DISCUSSION

We expand the interpretation of the above results with artifact-anchored technical detail that clarifies why the confirmatory test was uninformative for generation-level contamination and what concrete steps will increase audit sensitivity in future runs.

Execution factors that produced the degenerate paired result. Two execution facts recorded in the archived artifacts jointly explain the observed complete concordance (`n11 = 150`): (1) the confirmatory sample was drawn from the `deterministic_netops_task_generator_v5` generator and therefore all sampled items are provenance-stable synthetic tasks (each manifest entry carries the `source_identifier` prefix "synthetic-netops:"); and (2) the pipeline executed with `shared_initial_candidate` semantics (`independent_condition_generation = false`), so the executor reused a single model generation per item for both baseline and guarded episodes when no exclusions occurred. Both facts are recorded in the execution artifact set (notably `execution/task_manifest.jsonl` and `execution/model_configuration.json`) and the `deterministic_reconciliation` artifact confirms that `model_calls_used =`

150 rather than the preregistered two calls per item. Together these facts made the paired comparison insensitive to generation-level differences because baseline and guarded outputs were identical by construction absent a flag-triggered replacement.

Why per-item fuzzy overlap summaries are absent and the implications. The contamination detector implementation in the archived pipeline persisted detailed fuzzy-overlap outputs only when a tokenized 5-gram overlap exceeded the preregistered thresholds or when replacement handling was invoked. The `contamination_summary` artifact (`analysis/contamination_summary.csv`) contains no flagged rows for this execution; as a result, no global per-item overlap table was produced. This missing table is not an audit omission but a documented artifact of the detector workflow: the detector logged no threshold crossings and therefore the per-item score persistence path was not activated. The consequence is practical: readers cannot inspect a distribution of overlap distances to the 0.80 threshold for these items from the archive. This absence reduces post-hoc interpretability about how close items were to being flagged (e.g., whether maximum overlaps were 0.10 vs 0.60) and motivates a concrete workflow change—persist all per-item overlap scores regardless of threshold crossings—so future audits can report min/median/75th/ max overlap statistics and assess threshold sensitivity directly.

Sensitivity, sample size, and detectability tradeoffs (artifact-grounded). The preregistration documented power guidance tying detectable paired differences to the number of independent pairs and cross-model replication. The executed design used one shared generation per item (`model_calls_used = 150`) rather than the two independent generations per item in the maximal preregistered plan (up to 300 calls). Reducing independent trials reduces statistical power to detect modest paired differences: with fewer independent generation events the probability of observing discordant pairs (`n10` or `n01`) falls even if contamination is present in a non-deterministic or paraphrase-sensitive manner. The archived `paired_bootstrap` CI and exact McNemar results (95% CI [0.0,0.0];  $p = 1.0$ ) accurately reflect this degenerate sample: when every pair is concordant the paired estimand is exactly zero and statistical tests correctly report no evidence of change. Practically, audits aimed at detecting contamination in web-sourced benchmarks should increase independent generation counts (set `independent_condition_generation = true`), sample provenance-rich public items, and expand to multi-model replication to raise the expected discordance rate and ensure adequate power for targeted effect sizes.

Design changes supported by archive evidence. Based on the recorded execution and preregistered analysis plan, the following modifications are directly supported by the archived artifacts and address the reviewers' concerns: (a) sample public items with explicit provenance metadata (URLs/DOIs and timestamps) so provenance heuristics can meaningfully discriminate pretraining-era evidence; (b) set `independent_condition_generation = true` (recordable in `model_configuration.json`) to ensure baseline and guarded

outputs are independently realized, enabling observation of generation-level discordance; (c) persist per-item fuzzy 5-gram overlap scores in `analysis/contamination_scores.csv` (new artifact) regardless of threshold crossings so full overlap distributions are available; (d) expand to multiple model instances and independent seeds (increasing `model_calls_used` and `complete_pair_count`) to reveal cross-model signal and improve power; and (e) add paraphrase-robust detectors and human adjudication for borderline cases, because conservative lexical detectors can miss paraphrased memorization. Each recommendation follows from explicit gaps visible in the current archived outputs: `model_calls_used = 150` (execution manifest), an empty `contamination_summary.csv`, and the lack of a per-item overlap table in `analysis` artifacts.

Operational implications and caution about generalization. The artifact-verified result demonstrates that validator-backed correctness checks remain essential: `deterministic_netops_validator_v5` produced granular traces confirming functional correctness on all executed episodes. However, auditors and practitioners must be cautious about extrapolating a synthetic, shared-generation null to public benchmarks. The archival record supports only the narrow inference that, under provenance-stable synthetic sampling and shared generation semantics, the preregistered conservative detectors produced no high-confidence flags and the guarded policy made no changes. It does not address paraphrase leakage, web-sourced contamination, or multi-model behavior. For operational NetOps risk assessments, audits should therefore combine provenance-rich sampling, independent generations, cross-model checks, and human adjudication to obtain evidence that generalizes to production corpora.

Summary and concrete next steps. The expanded artifact diagnosis clarifies the precise mechanism of degeneracy and prescribes artifact-anchored corrective steps for future audits. Importantly, these recommendations are implementable within the same execution framework: changing a single boolean (`independent_condition_generation`) and enabling full per-item score persistence produce the necessary artifacts to evaluate contamination sensitivity. Future preregistered runs that implement these changes will yield non-degenerate paired comparisons when contamination or templated leakage exists in the sampled items; the present archived run documents the baseline mechanics and the specific gaps that must be closed to make a confirmatory paired contamination audit informative for public benchmarks.

## VI. LIMITATIONS

- Single-model execution (`gpt-5-mini`) — results do not generalize across other LLM families, sizes, or training corpora.
- Sampled items were provenance-stable synthetic/deterministically generated tasks (source identifier prefix `"synthetic-netops:"`) for this run; absence of high-confidence flags here may not reflect contamination prevalence in real-world, web-crawled benchmarks.

- Generation semantics used `shared_initial_candidate` across baseline and guarded conditions (`independent_condition_generation = false`), producing identical model outputs when no exclusions occurred and thus limiting sensitivity to interventions that rely on independent re-generation.
- Contamination detection relied on preregistered conservative heuristics (exact match and fuzzy 5-gram overlap thresholds); subtler contamination patterns (paraphrased memorization, partial leakage) may escape these heuristics and require richer provenance or human adjudication.
- Passing the `deterministic_netops_validator_v5` indicates satisfaction of the checked functional constraints but does not imply comprehensive production readiness or absence of semantic regressions outside the validator’s scope.

## VII. CONCLUSION

We executed a fully preregistered, artifact-anchored paired audit of conservative contamination detectors for LLM-for-NetOps evaluation. In the reproducible 150-pair synthetic sample we observed zero preregistered high-confidence contamination flags and identical validator pass rates (150/150 baseline, 150/150 guarded). The degenerate null outcome is artifact-supported and stems from provenance-stable synthetic sampling and shared generation semantics; it does not imply absence of contamination in web-sourced benchmarks or failure of the preregistered detectors in different sampling regimes. We provide an explicit preregistration→execution reconciliation, confirm deterministic reconciliation checks passed, and recommend concrete design changes—provenance-rich sampling, independent generation per condition, multi-model scope, paraphrase-sensitive detectors, and matched power calculations—to increase audit sensitivity in future studies. All experiment artifacts, validator traces, analysis inputs/outputs, and execution manifests are archived and available as recorded in the Disclosure Statement below.

## DISCLOSURE STATEMENT

This study was executed under the autonomous-run preregistration `contamination-audit-netops-2026-v1` (preregistration SHA-256 recorded in the archived bundle). LLM used: `gpt-5-mini` (`declared_model_version = "gpt-5-mini"`). Orchestration adapter: `hosted_netops_gvr_v1`. Deterministic task generator: `deterministic_netops_task_generator_v5` (version 5.0). Deterministic validator: `deterministic_netops_validator_v5` (version 5.0). Representative executed artifacts (by path) include `execution/model_configuration.jsonl`, `execution/task_manifest.jsonl`, `execution/raw_results.jsonl`, `analysis/paired_contingency_table.csv`, and `analysis/contamination_summary.csv`; the full execution bundle contains raw responses, validator traces, execution logs, and the transformation manifest. Exact immutable initial master-prompt reference: `provenance/master_prompt.txt` (SHA-256: `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba`). Topic constraints (Generative

AI and LLMs for NetOps) were selected by the human Research Director prior to the autonomy lock. After the autonomy lock, the AI system autonomously performed literature search, gap selection, preregistration, experiment execution, analysis, manuscript drafting, peer-review simulation, and revision. All generation prompts, model outputs, validator traces, sampling seeds, replacement rules, execution logs, and analysis artifacts were produced and archived by the autonomous pipeline and are included in the execution bundle. No manual human editing of technical content, analyses, references, figures, tables, captions, or the Disclosure Statement occurred after the autonomy lock. Pre-lock development and rehearsal runs occurred (permitted) but pre-lock artifacts were not used as empirical evidence for this final study unless re-created under the locked pipeline. The execution followed preregistered missingness, retry, and exclusion policies; no deviations occurred.

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